

Aptimizer: Complete Feature List (V1 to X+)

A consolidated feature index across all eight versions. 420 features total, of which 324 are built and 28 are partially built.

Version	Focus	Features	Built	Status
Aptimizer V1	Core Civil Engineering Planning Platform	124	124	Complete
Aptimizer V2	GIS & Site Intelligence	27	25	In progress
Aptimizer V2.5	Precision, Programme & Assurance	114	113	In progress
Aptimizer V3	AI Engineering Platform	55	55	Complete
Aptimizer V4	Enterprise Collaboration	13	7	In progress
Aptimizer V5	BIM & Smart Construction	13	0	Planned
Aptimizer X	AI Civil Engineering Operating System	55	0	Early
Aptimizer X+	Future Vision	19	0	Early

■ Built and running

■ Partially built

■ Planned

Build Status: Where Aptimizer Stands Today

Aptimizer is built and running through **V3**. V1, V2 and V3 are functionally complete, and the V2.5 layer added since the last feature list is complete apart from one item. V4 is roughly half built. V5, X and X+ are not built.

The headline figure: **324 of 420 catalogued features are built**, with a further 28 partially built. Everything still outstanding sits in enterprise workflow, BIM interoperability, and the long-range operating-system vision.

What was added since the previous feature list

The previous list catalogued 253 features across seven versions. This revision adds 167, almost all of them in a new **V2.5** band that did not exist before. The largest additions are:

- **Development controls and setbacks.** Per-edge inward setbacks, setback minimums derived from plot area, the height-driven open space table, and the height versus road width limit, with verification and warnings.
- **A real site layout engine.** Buildable envelope generation, road network reservation with ring roads and internal spines, amenity block placement, greedy grid tower packing with rotation sweep and spacing enforcement, and a hard boundary containment guarantee.
- **Vastu and generative architecture.** Sector mapping, sector-anchored unit packing, per-unit Vastu audit, MEP position audit, and a measured plan realism check against a built reference drawing.
- **Programme and scheduling.** Activity generation from quantities, full CPM with forward and backward passes, critical path, float verification, work calendars, crew efficiency, prop removal safety rules, a target date solver, cash flow, resource histogram and line of balance.
- **Development finance.** Saleable area derivation, S-curve cost spread, monthly cash flow, NPV, IRR and break-even.
- **Data reliability.** Nine per-module data checks, freshness tracking, cross-module consistency checks, compliance margin reporting and an overall trust score.
- **IS and NBC code intelligence.** A 74-entry clause registry, clause text corpus indexing, semantic retrieval, and verification of every clause the model cites against the real registry.

What is not built

- Genetic refinement of tower placement. Placement is greedy grid packing only. The fitness function and GA configuration exist; the refinement stage does not.
- Approval workflow, comments and reviews, and workflow automation in V4.
- All BIM interoperability in V5. There is no IFC, DWG, DXF, Revit or AutoCAD export.
- Digital twin, IoT and construction monitoring in V5.
- The Aptimizer X operating system layer. Several X features exist in partial form because a V2.5 or V3 module already covers part of the ground, but none is complete at X scope.

Stated plainly: Aptimizer is a complete V3 product with a substantial V2.5 precision layer on top, partial V4 collaboration, and no V5 or X capability yet.

How the Platform Is Used: Workflow Order

The feature index that follows is ordered by version, because its job is to show what was built when and what comes next. The platform itself is not used in that order. A project moves through five stages, and each stage draws on features from several versions at once. This page maps one onto the other.

Stage	Workspace Module	Draws On	Built
1. Site	Plot & Setbacks	V1, V2.5	Yes
	GIS Intelligence	V2	Yes
2. Design	Apartment Planning	V1, V2.5	Yes
	Parking	V1	Yes
	3D Visualisation	V2.5	Yes
3. Engineering	Calculations	V1	Yes
	IS/NBC Engineering	V2.5	Yes
	BOQ & Quantities	V1, V2.5	Yes
4. Cost & Programme	Cost Estimation	V1	Yes
	Programme	V2.5	Yes
	Feasibility & ROI	V2.5	Yes
5. Deliver	Compliance	V1, V2.5	Yes
	Data Reliability	V2.5	Yes
	Reports	V1	Yes
	Versions & Team	V1, V4	Partial

Cutting across every stage

Two capabilities do not sit at a single point in the workflow, which is why they do not appear as a stage of their own.

■ **AI assistance (V3).** APT, the optimisers and the AI advisors are reachable from every module, and each draws on the values that module has already computed. The assistant is context-aware by stage rather than being a separate destination.

■ **Data reliability (V2.5).** The checks run continuously against every module, but the report is placed in Deliver on purpose. The question it answers is whether the drawing set can be issued, and that is worth asking immediately before the report is generated, not while the design is still moving.

Reading order note: a stage is only as trustworthy as the stage before it. Setbacks decide the buildable envelope, the envelope decides tower placement, placement decides areas, areas decide quantities, quantities decide cost and programme. An error at stage 1 propagates through all five, which is the reason the data reliability checks exist at all.

Aptimizer V1: Core Civil Engineering Planning Platform

Objective: Develop a platform for apartment planning, quantity estimation, cost estimation, and compliance calculations.

124 features | ■ 124 built | ■ 0 partial | ■ 0 planned

User & Project Management

- User Registration & Login
- User Profiles
- Project Dashboard
- Version Comparison
- Project Activity History
- Admin Console
- Role-Based Access Control
- Project Creation
- Project Versioning
- Project Sharing
- Public Compliance Link
- Autosave & Recompute

Plot Management

- Manual Plot Entry
- Plot Dimensions
- Plot Boundary Editing
- Road Access Definition
- Coordinate (Lat/Lng) Entry
- Polygon Plot Creation
- Plot Orientation
- Plot Area Calculation
- Road-Edge Classification
- Polygon Repair & Validation

Apartment Planning

- Apartment Type Selection
- Floor Planning
- Corridor Planning
- Lift Planning
- Common Area Planning
- Procedural Room Layout Generator
- Layout Validation & Retry
- Tower Planning
- Room Planning
- Staircase Planning
- Balcony Planning
- Unit Mix Definition
- Whole-Tower Floor Plate Generation

Civil Engineering Calculations

- Plot Area
- Built-up Area
- Ground Coverage
- FSI Calculation
- Density Calculation
- FAR Derivation Breakdown
- Carpet Area
- Super Built-up Area
- FAR Calculation
- Open Space Calculation
- Area Derivation Audit Trail

Parking Planning

- Basement Parking
- Visitor Parking
- Accessible Parking
- Parking Efficiency Analysis
- Carpet-Band State-Aware Norms
- Shared Site Parking Pool
- Ground Parking
- EV Parking
- Parking Capacity Calculator
- Ramp Planning
- Per-Tower Demand & Supply
- ECS Conversion

Utility Planning

- Water Demand
- Overhead Tank
- WTP Planning
- Electrical Room
- Underground Tank
- STP Planning
- Rainwater Harvesting
- Pump Room

Quantity Estimation

- Concrete
- Steel
- Sand
- Tiles
- Doors
- Plumbing
- Waterproofing
- Cement
- Bricks/Blocks
- Aggregate
- Paint
- Windows
- Electrical
- Finishing Materials

BOQ

- Automatic Bill of Quantities
- Labour Summary
- BOQ PDF Export
- Structure-Derived Quantity Take-off
- Mix Proportion Derivation
- Material Summary
- Equipment Summary
- BOQ Excel Export
- Column & Beam Section Sizing
- Take-off Sanity Checks

Cost Estimation

- Material Cost
- Equipment Cost
- Cost per Flat
- Budget Summary
- Preliminaries
- Contingency
- GST
- Labour Cost
- Construction Cost
- Cost per sq m
- Wastage Allowance
- Overhead & Profit
- Escalation

Compliance Engine

- FAR Validation
- Stair Width
- Lift Compliance
- Accessibility Checks
- Height vs Road Width Rule
- Clause-Referenced Results
- FSI Validation
- Corridor Width
- Fire Exit Checks
- Setback Validation
- Open Space vs Height Rule
- Compliance Margin Warnings

Reports

- BOQ Report
- Quantity Report
- Compliance Report
- PDF Export
- Excel Workbook Export
- Cost Report
- Parking Report
- Executive Summary
- Per-Tower Report Sections

Aptimizer V2: GIS & Site Intelligence

Objective: Integrate maps, satellite imagery, and geospatial analysis.

27 features | ■ 25 built | ■ 1 partial | ■ 1 planned

GIS & Site Intelligence

- | | |
|-------------------------------|-------------------------------|
| ■ OpenStreetMap Support | ■ GIS Coordinate Input |
| ■ Polygon Drawing | ■ Plot Boundary Detection |
| ■ Existing Building Detection | ■ Road Detection |
| ■ Road Width Detection | ■ Green Area Detection |
| ■ Water Body Detection | ■ Terrain Analysis |
| ■ Elevation Analysis | ■ Slope Analysis |
| ■ Flood Risk Analysis | ■ Wind Analysis |
| ■ Sun Path Analysis | ■ Solar Position & Sun Events |
| ■ Annual Insolation | ■ Solar Potential |
| ■ Transit Proximity Analysis | ■ Accessibility Analysis |
| ■ Site Suitability Score | ■ Infrastructure Analysis |
| ■ Buildability Analysis | ■ AI Site Analysis |
| ■ Overpass Feature Caching | ■ Google Maps Integration |
| ■ Satellite Imagery Basemap | |

Aptimizer V2.5: Precision, Programme & Assurance

Objective: Make every number defensible: real development controls, a real site layout engine, a real programme, and a stated confidence in the data behind them.

114 features | ■ 113 built | ■ 0 partial | ■ 1 planned

Development Controls & Setbacks

- Per-Edge Inward Setback Application
- Height-Driven Open Space Table
- Recommended Controls Engine
- Development Control Overrides
- Setback Minimums from Plot Area
- Height vs Road Width Limit
- Setback Verification & Warnings
- Yield Recommendation

Site Layout Engine

- Buildable Envelope Generation
- Local Tangent-Plane Projection
- Perimeter Ring Roads
- Straight-Line Corridor Geometry
- Amenity Block Placement
- Greedy Grid Tower Packing
- Inter-Tower Spacing Enforcement
- Boundary Containment Guarantee
- Layout Recommendation Endpoint
- Genetic Refinement of Placement
- Polygon Repair & Cleaning
- Road Network Reservation
- Internal Spine & Driveways
- Largest Inscribed Block Detection
- Green & Open Space Reservation
- Footprint & Rotation Sweep
- Tower Count & Height Caps
- Shared Fitness Scoring
- Engine Versioning & Signatures

Vastu & Generative Architecture

- Vastu Sector Mapping
- Unit Programme Generation
- MEP Position Audit
- Vastu-Aligned Floor Plates
- Plan Realism Check
- Sector-Anchored Unit Packing
- Vastu Audit per Unit
- Sector Report
- Architectural Template Generation
- Room Aspect & Dimension Checks

Programme & Scheduling

- Activity Generation from Quantities
- Critical Path Identification
- Float Verification
- Duration from Output Rates
- Wage Premium Modelling
- Schedule Safety Audit
- Task Overrides & Edits
- Cash Flow Curve
- Line of Balance
- Gantt Visualisation
- CPM Forward & Backward Pass
- Total & Free Float
- Work Calendar & Holidays
- Crew Efficiency Modelling
- Prop Removal Safety Rules
- Scheduled Date Audit
- Target Date Solver
- Resource Histogram
- Milestones

Development Finance

- Saleable Area Derivation
- S-Curve Cost Spread
- NPV
- Break-even Point
- Finance Configuration
- Revenue Model
- Monthly Cash Flow
- IRR
- Preliminaries per Month

Data Reliability & Assurance

- Plot Data Check
- Tower Data Check
- GIS Data Check
- Engineering Check
- Programme Check
- Cross-Module Consistency Checks
- Module Warning Rollup
- Data Health Report Export
- Development Controls Check
- Site Layout Check
- Parking Check
- Finance Check
- Data Freshness Tracking
- Compliance Margin Reporting
- Overall Trust Score

IS & NBC Code Intelligence

- Central Code Constant Registry
- Clause Text Corpus Ingestion
- Semantic Clause Retrieval (RAG)
- Code Question Answering
- Citation Verification Against Registry
- 74-Entry Clause Registry
- Clause Chunking & Indexing
- Code Search Endpoint
- Citation Extraction
- Cited Extracts in Answers

Engineering Modules (IS/NBC)

- Structural Load Estimator (IS 875)
- Seismic Zone & Base Shear (IS 1893)
- Concrete Mix Design (IS 10262)
- Storm Water & RWH (IS 3764)
- Fire Safety Compliance (NBC Part 4)
- Green Building Preliminary Rating
- Embodied Carbon
- Wind Force Coefficients (IS 875-3)
- Foundation Advisor (IS 6403/1904)
- Water Infrastructure (IS 1172)
- Parking Compliance (NBC/SP:21)
- Accessibility Compliance (NBC Part 3)
- Column Grid Optimizer
- Plantation Plan

3D & Visualisation

- 3D Site Model
- Road & Plot Rendering
- Central Park & Landscape
- Site Diagrams
- Live Metrics Strip
- Tower Massing
- Amenity & Clubhouse Rendering
- Floor Plate Viewer
- Blueprint View
- Command Palette

Aptimizer V3: AI Engineering Platform

Objective: Introduce AI-driven planning and decision support.

55 features | ■ 55 built | ■ 0 partial | ■ 0 planned

AI Planner

- Automatic Layout Generation
- Apartment Mix Optimisation
- Open Space Optimisation
- Smart Tower Placement
- Floor Optimisation

AI Civil Engineer

- Civil Design Suggestions
- FSI Optimisation
- Parking Optimisation
- FAR Optimisation
- Utility Optimisation

AI Structural Advisor

- Column Grid Suggestions
- Foundation Recommendations
- Beam Layout Suggestions
- Structural Risk Warnings

AI Quantity Surveyor

- Quantity Optimisation
- Steel Cutting Optimisation
- BOQ Optimisation
- Waste Reduction
- Tile Waste Optimisation

AI Cost Engineer

- Cost Prediction
- Cost Comparison
- Grade Sweep Analysis
- Budget Optimisation
- Cost Saving Suggestions

AI Sustainability

- Green Building Score
- Solar Potential
- Tree Plantation Suggestions
- Carbon Footprint
- Water Efficiency

AI Business Consultant

- ROI Analysis
- Project Feasibility
- Profit Forecast
- Break-even Analysis

AI Recommendation Engine

- Design Improvements
- Cost Optimisation
- Engineering Suggestions
- Material Recommendations

APT: In-App AI Assistant

- Project Q&A;
- Report Explanation
- Streaming Responses
- Project State Context Builder
- Query Classification
- Chat Threads & History
- Provider Fallback & Retry

Scenario Comparison

- Compare Layouts
- Compare Parking
- Compare ROI

- Civil Engineering Guidance
- Scenario Comparison
- Module-Aware Opening Questions
- Context & Analysis Caching
- Capability Questions
- Citation Verification

- Compare Costs
- Compare Sustainability

Aptimizer V4: Enterprise Collaboration

Objective: Enable teamwork and enterprise workflows.

13 features | ■ 7 built | ■ 3 partial | ■ 3 planned

Collaboration & Workflow

- | | |
|----------------------------|----------------------|
| ■ Multi-user Collaboration | ■ Project Sharing |
| ■ Team Roles | ■ Version Control |
| ■ Activity Logs | ■ Task Assignment |
| ■ Approval Workflow | ■ Comments & Reviews |
| ■ Workflow Automation | |

Engineering Decision Support

- | | |
|-------------------------|----------------------------|
| ■ Construction Timeline | ■ Labour Planning |
| ■ Equipment Planning | ■ Utility Network Planning |

Aptimizer V5: BIM & Smart Construction

Objective: Prepare for BIM, Digital Twin, and construction monitoring.

13 features | ■ 0 built | ■ 0 partial | ■ 13 planned

BIM Readiness

- | | |
|---|---|
| ■ IFC Export | ■ DWG Export |
| ■ DXF Export | ■ Revit Compatibility |
| ■ AutoCAD Compatibility | |

Digital Twin Preparation

- | | |
|--|--|
| ■ IoT Readiness | ■ Smart Sensor Integration |
| ■ Progress Tracking | ■ Quality Monitoring |
| ■ Safety Monitoring | ■ Delay Prediction |
| ■ Predictive Maintenance | ■ Facility Management |

Optimizer X: AI Civil Engineering Operating System

Objective: A unified AI-powered platform for the entire project lifecycle.

55 features | ■ 0 built | ■ 22 partial | ■ 33 planned

AI Operating System

- AI Engineering Copilot
- Engineering Knowledge Graph
- Engineering Decision Log
- AI Engineering Consultant
- Engineering Memory
- Explainable AI

Intelligent Planning

- One-Click Project Generation
- AI Township Planner
- Apartment Capacity Prediction
- Smart Building Recommendations
- AI Layout Generator
- Mixed-Use Development Planning
- Land Suitability Analysis

GIS & Urban Intelligence

- City Growth Prediction
- Infrastructure Growth Prediction
- Disaster Risk Analysis
- Flood Analysis
- Noise & Pollution Analysis
- Land Value Prediction
- Climate Analysis
- Earthquake Analysis
- Wind Analysis

Smart Procurement

- Live Material Prices
- Supplier Intelligence
- Inventory Planning
- Material Price Forecasting
- Procurement Calendar

Financial Intelligence

- Cash Flow Planning
- Investment Analysis
- Profitability Dashboard
- Lifecycle Cost Analysis
- ROI Forecasting

Sustainability

- Green Building Planner
- Solar Energy Planning
- ESG Reporting
- Carbon Emission Calculator
- Water Conservation Analysis

Construction Intelligence

- Construction Timeline Optimisation
- Labour Optimisation
- Construction Delay Prediction
- Resource Allocation
- Equipment Scheduling
- Risk Prediction

Reports & Decision Support

- AI Report Writer
- Interactive Analytics
- Scenario Simulation
- Executive Dashboards
- Decision Sandbox
- Engineering Benchmarking

Marketplace & Ecosystem

- Engineering Marketplace
- API Marketplace
- Government Approval Assistant
- Plugin Marketplace
- Educational Mode
- Tender Document Generator

Aptimizer X+: Future Vision

Objective: A future roadmap beyond Aptimizer X.

19 features | 0 built | 2 partial | 17 planned

Autonomous AI Engineering

- Conversational Project Design
- Autonomous Compliance Checking
- AI Negotiation with Suppliers
- Multi-Agent Engineering Teams
- Autonomous BOQ & Costing

Generative Design

- AI-Generated 2D Floor Plans
- Facade Design Suggestions
- Parking Layout Generation
- AI-Generated 3D Building Concepts
- Landscape Design

Smart City Platform

- City-Scale Planning
- Utility Network Optimisation
- Infrastructure Demand Forecasting
- Traffic Simulation
- Urban Digital Twin

XR & Robotics

- AR Site Visualisation
- Drone Survey Integration
- VR Project Walkthroughs
- Robotics-Assisted Construction Planning